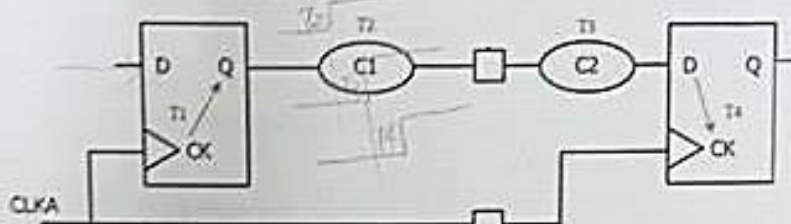


NCTU-EE IC LAB – SPRING 2016

Final Online Test - Quiz 1:30p.m. ~ 3:00p.m.

1. Please specify the following by using T1, T2, T3, T4. (12pts)

- (1) Clock cycle
- (2) Input delay
- (3) Output delay



2. Please fill in the blanks below to finish the commands for multicycle(8pts)

set_multicycle_path (?) -from FF_4/CK -to FF_5/D
set_multicycle_path (?) -hold-end -from FF_4/CK -to FF_5/D



3. Please draw the waveform to explain the following two assertions. (10pts)

```
property p_req_ack;
  @(posedge clk) mem_en |-> (req ##2 ack);
endproperty: p_req_ack
```

```
property p_req_ack;
  !@(posedge clk) mem_en |> (req ##2 ack);
endproperty: p_req_ack
```

1.

$$(1) T_1 + T_2 + T_3 + T_4$$

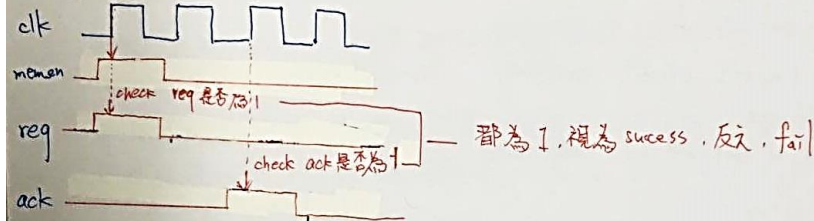
$$(2) T_1$$

$$(3) T_4$$

2. 5, 4

① check mem_en 是否為 1, 等於 0 則視為 success, 等於 1 則必須斷之後的 sequence 到

3.



4.

lower $V_{DD} \Rightarrow$ lower (switching current) (Internal current) - leakage current
higher $V_{th} \Rightarrow$ lower leakage current (multi V_{th})

power gating $\Rightarrow$ 當電路不需使用時, 將電源跟電路斷開, 就不會有功耗

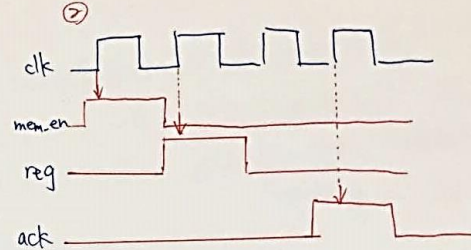
clock gating $\Rightarrow$ 避免不必要的 signal transition (0 $\rightarrow$ 1, 1 $\rightarrow$ 0), lower switching current

3. IR drop $\Rightarrow$ 因為 wire 本身有電阻, 所以 V_{DD} 或 V_{SS} 的電壓會因為 wire 的電阻而使得 Chip 某些位置 cell 的 V_{DD} 下降, V_{SS} 上升

fix: add more power stripes. (IR)

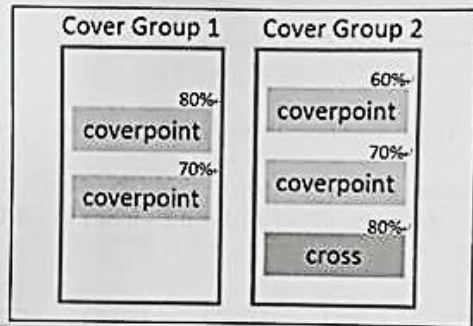
① performance: 因為 IR drop 會使得 V_{DD} 下降, current 下降, delay 上升, spec 可能無法達到

② setup time violate hold time violate: $\because$ delay 上升, 所以可能無法符合 setup time, hold time 要求



- Please list four ways to reduce the power consumption, and briefly explain why. (12pts)
- Please explain what is IR drop? And give two examples that how IR drop influences on design) (10pts)
- Explain antenna effect, and give two method to fix it. (10pts)
- What is the purpose of placement blockage? How and where do we set this blockage? (10pts)
- Calculate the coverage of the whole design, the coverage of coverpoints or cross is specified as shown on the picture (10pts)

Design

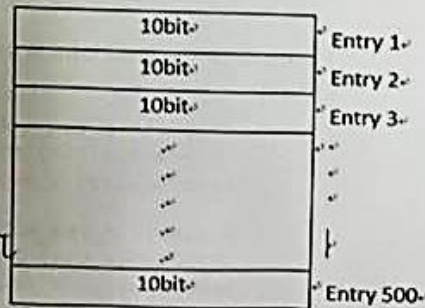


60%



- Please explain why we use the latch on the clock gating. (Please also draw the or-clock gating with latch and the and-clock gating with latch) (10pts)

- Please design a memory with following spec: (8pts)
 - Number of words: 500
 - Number of bits: 10



- How many input pins are needed (at least)? data bus? bit
- How many bits of address are needed (at least)?

- antenna effect: 在蝕刻時, 不需要的金屬會被蝕刻掉, 而剩下的部分金屬會累積電子, 而當累積電子過多, 會使得電壓上升, 打穿 gate oxide, 使得 MOSFET 壞掉, 通常 wire 接太長, 較容易發生 antenna effect.

- add antenna cell (diode), 使電子流掉
 - add jumper, 使 wire 不要太長, 電子累積降低

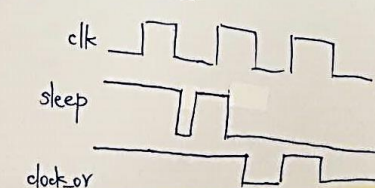
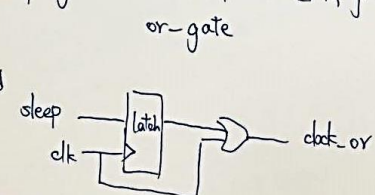
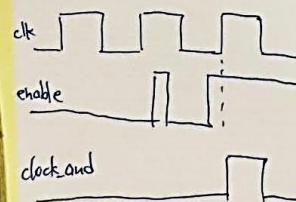
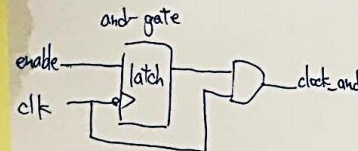
- placement blockage: 使得在放 standard cell 時, 不會擺放到 power stripes 的那幾層 metal layer 避免再擺放 macro or standard cell 到我們所限制的區域內

- 當我們有用到 macro 時, 會希望 macro 周圍不要放 cell, 避免太擁擠, 就會設定 macro 周圍區域 不要再擺放 cell
 - 當要擺放 standard cell 時, 會希望不要直接連到 power stripes, 所以會設定 power stripes 的那幾層 metal 不能放 standard cell

$$8. \frac{\left(\frac{(80+70)}{2} + \frac{(60+70+80)}{3}\right)}{2} = \frac{75+70}{2} = 72.5\%$$

$$10. \textcircled{1} 10 \textcircled{2} 9 (2^{9.512})$$

- 因為如果有 glitch 產生, 會有無法預期的結果發生
 - 利用 latch 可以使得 glitch 被濾掉, 不會有 glitch 發生



or-gate is better because the first latch doesn't work